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A Hybrid Computer Vision and Bioinformatics Framework for Detecting Oncogenic Mutations in Human Genomic Data

Ezra Chaitanya Prasad G Y
Computational Intelligence
SRMIST University
Chennai,India
eg2465@srmist.edu.in

P Harrish Jagathroy
Computational Intelligence
SRMIST University
Chennia,India
hp3581@srmist.edu.in

Dr.Sheryl Oliver A
Assistant Professor
Computational Intelligence
SRMIST University
Chennai,India
sheryloa@srmist.edu.in

Dr.M.Kiruthika
Assistant Professor
Computational Intelligence
SRMIST University
Chennai,India
kiruthim4@srmist.edu.in

***Abstract*— one of the most complex challenges in modern medicine is cancer, being mostly driven by somatic mutations that affect cellular regulation. Techniques like BLAST and FASTA are primarily the most seemed sequence alignment algorithms for identifying such mutations. Although very convenient for small scale analysis, these also face high scalability bottlenecks, high computational latency and memory costs when the whole genome sequencing data occurs. Most importantly, global differences in structure and nonlinear correlations in complex genomic signatures are failed to be captured by linear alignment methods.**

The biggest hurdle with finding cancer accurately depends on identifying which somatic mutations in the whole genome sequence is causing the tumors.in addition, alignment based on conventional approaches necessitate extended timelines for resolution and fail to detect long range gene level interactions. This project solely intends to design a very specific deep learning system that doesn't require alignment and can classify the somatic mutations from DNA directly, by analysing the text sequences. We have created a hybrid architecture that comprises of a one-dimensional convolutional neural network for figuring out the local motifs, a Bidirectional Gated Recurrent Unit for modelling sequential context, and a Squeeze-and-Excitation block for specifying certain features. This model was trained using the COSMIC (Catalogue of somatic mutations in cancer) dataset and GRCh38 human reference genome. It is dedicated for the 101-base pair windows. The human genetic data variations is balanced out by additionally applying reverse complement and dynamic class weighing augmentations.

Keywords*— *Bioinformatics, Deep Learning, 1D-CNN, BiGRU, Squeeze-and-Excitation, Somatic Mutation, COSMIC, Computational Biology

I. Introduction

Larger quantities of genetic and epigenetic changes causes cells to be incapable of controlling themselves which serves as the primary reason for cancer.in biological terms, these somatic mutations take control of normal bodily functions and make cells resist apoptosis, continue to split and divide for longer period and then initiating angiogenesis. The tumors develop resistance to immunological and therapeutic stressors over time, causing internal changes in their structures over time. This is exactly how it makes medicines inefficient and causes treatment plans to be unreliable and tougher to plan INITIALLY.

On top of that, the consequence of the biology reaches far beyond the main tumour, an example being around 90 percent of deaths related to cancer occur because of metastasis (cancer cells reaching distant organs). This could be a clinical imperative instead of academic ambition to understand how exactly which molecules make up cancer because the inconsistent behaviour is written into the tumour's DNA. When specific actionable mutations are identified (BRCA1/2 in breast cancer), it allows accurate deployment of targeted therapies and precision oncology.

Now the main method used in current clinical diagnostics to figure out this complicated mutational landscape is next generation sequencing (NGS) . NGS makes it possible to sequence millions of DNA fragments at the same time. This helps doctors detect insertions/deletions (indels), single nucleotide variants (SNVs), and bigger structural changes.

However, when sequence depths go up, the precision of the approach making Mutect2 so accurate also causes large processing difficulties. Whole Genome Sequencing (WGS) provides a lot of data, which makes it difficult for regular variant callers to operate well because they were made for CPU-bound frameworks.
There are several structural bottlenecks:

1. **Big I/O Problems**: Molecular biology pipelines need to process, categorize, and generate big Binary Alignment Map (BAM) or CRAM files. Transferring these files in conventional memory often takes down the entire computing cluster simply due to the disk I/O necessities.
2. **The Burden of Local Assembly**: Mutect2 employs graphs to detect "active regions" and then puts the reads back together in the same place to find complex indels. This method needs a lot of computing power. This graph construction gets increasingly intricate quite quickly, which makes the calculations take a lot longer. This is because clinical oncology often demands high-depth sequencing (for example, 100x to 500x coverage) to discover unusual subclones.

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3. **Memory Use**: There aren't many simultaneous operations that can run on standard nodes without causing issues with memory. This is because modern de Bruijn graph layouts for haplotype assembly depend on a lot of RAM to function.
4. **Prohibitive Wall-Clock Time:** Due to these sequential, single-threaded limitations in active regions, it might require hundreds of computing hours to process a single extensive normal - tumour pair into a normal Mutect2 pipeline.

Computational biology is now at the forefront of precision oncology thanks to the development of Next-Generation Sequencing (NGS), which has made whole-genome data more accessible. Somatic mutations, which are post-zygotic changes in the DNA sequence that interfere with regulatory networks and cellular proliferation pathways, are the primary cause of cancer. It is still very difficult to distinguish these harmful driver mutations from benign passenger mutations or sequencing errors.

The Burrows-Wheeler Transform (BWT) and other sequence alignment techniques are used in conventional variant calling procedures. Despite their robustness, these techniques are computationally demanding and frequently fall short of capturing the intricate, non-linear epistatic interactions that are a feature of genetic language. We suggest an alignment-free Deep Learning framework to close this gap. We use a hybrid 1D-CNN and Bidirectional GRU (BiGRU) architecture enhanced with Squeeze-and-Excitation (SE) attention methods to categorize somatic mutations directly from raw 101-bp nucleotide sequences by treating the genome as a sequential vocabulary.

II. Related Work

The constant requirement to distinguish genuine biological signals from overwhelming genomic noise has propelled the computational development of variant calling and sequence analysis. This development can be categorized historically into three separate periods: the use of shallow machine learning for feature engineering, the adoption of deep convolutional models, and the current trend toward attention-guided recurrent architectures.

A. The Limits of Feature Engineering: Support Vector Machines in Genomics

Shallow machine learning, particularly Support Vector Machines (SVMs), was a major component of early attempts to improve upon the strict statistical priors of conventional variant callers prior to the supremacy of deep neural networks. In order to map complex sequencing data into a high-dimensional feature space and find the best hyperplane to distinguish real variants from sequencing artifacts, SVMs were widely used in tools intended to filter somatic mutations or categorize variant pathogenicity.

SVMs gave bioinformaticians a clear edge over fixed-threshold filtering by enabling them to build multi-dimensional models based on read features. To detect systemic aberrations that conventional Bayesian models might mistakenly categorize as high-quality Single Nucleotide Polymorphisms (SNPs), for example, an SVM could be trained using a vector of hand-crafted features, such as read depth, mapping quality, base quality, allele frequency, and strand bias.

However, the "curse of dimensionality" and the rigid dependence on feature engineering severely limit the effectiveness of SVMs in genomics. In a very subjective and dataset-dependent process, human specialists had to manually identify, extract, and choose which genetic traits were significant. SVMs also handle input data as discrete, tabular points. They are unable to learn de novo motifs or comprehend the larger biological context around a mutation because they completely ignore the spatial, sequential character of DNA.

B. The Deep Learning Paradigm Shift: DeepSEA and DanQ

Deep learning in genomics eventually solved the problems with manual feature extraction by enabling models natively generate hierarchical representations from raw DNA sequences.

At that time, DeepSEA (2015) was a groundbreaking framework. DeepSEA showed that a regular Convolutional Neural Network (CNN) could predict the chromatin effects of changes in the sequence with single-nucleotide accuracy. DeepSEA used a three-layer CNN to construct a chain of automated motif scanners by treating one-hot encoded DNA sequences like 1D pictures. The convolutional filters might discover short, recurring patterns in positions, like position weight matrices, without any biological labels.

DanQ (2016) tackled the problem by implementing a hybrid architecture that greatly enhanced genomic deep learning. DanQ passed the feature maps directly into a Bi-directional Long Short-Term Memory (Bi-LSTM) recurrent neural network after using a single convolutional layer to find local motifs.

Adding the Bi-LSTM layer changed everything. To capture the long-range dependencies and physical limitations between the motifs revealed by the CNN, the network could read the sequence in both the forward and reverse-complement orientations. DanQ came with a large computational cost, even though it showed significant performance improvements over DeepSEA (sometimes more than a 50% relative improvement in precision-recall metrics for certain regulatory markers). Because of how complicated its internal processes are (input, output, and forget gates), LSTMs are known for needing a lot of parameters. When DanQ's Bi-LSTM layer is used with a lot of data, like in high-depth clinical sequencing, the memory and training time become too much.

C. The Superiority of BiGRU+SE Networks

We propose a distinctive hybrid architecture that integrates a Bidirectional Gated Recurrent Unit (BiGRU) with a Squeeze-and-Excitation (SE) attention mechanism to mitigate the computational constraints of DanQ while enhancing its responsiveness to significant illness changes. Our method has two significant structural benefits over typical deep learning models in genomics.

Computational Efficiency via BiGRU: LSTMs are too complicated for many genomic applications, even though they are important for capturing the bidirectional "grammar" of DNA. We switch out the LSTM for a BiGRU. The forget and input gates are combined into one "update gate" in GRUs, which cuts down on the number of parameters and matrix multiplications needed during backpropagation. The BiGRU does the same thing as DanQ: it models the long-range spatial distances between genomic motifs. However, it converges much faster and uses much less memory. This level of computing performance is necessary for variant calling algorithms to be able to handle whole-genome sequencing (WGS) data in a reasonable amount of time for therapeutic use.

Dynamic Feature Recalibration via Squeeze-and-Excitation (SE): The primary deficiency of both DeepSEA and DanQ lies in their equal treatment of all identified motifs; their convolutional layers transmit all feature maps to the recurrent layers without establishing any hierarchy. Our architecture fixes this by putting a SE network block just after the motif-scanning convolutions.

The SE block brings in the channel-wise attention mechanism. At first, it "squeezes" the global spatial information of the genomic sequence into a short channel description. After that, it learns a set of adaptive weights for each channel (motif), which "excites" the network. This means that the network can change its focus on the fly in terms of biology. If the network finds a highly informative mutational signature that is unique to cancer, the SE block increases that feature channel. On the other hand, it intentionally blocks feature channels that are related to known sequencing artifacts or background genomic noise.

The BiGRU+SE architecture is far better than early hybrid models because it combines the adaptive focus of a SE attention block with the lightweight, long-range contextual awareness of a BiGRU. It adds a dynamic, attention-guided filtering system that is far better at dealing with the noise that comes with clinical sequencing data, while yet keeping the "grammatical" knowledge of DNA sequences.

Early methods for classifying variants used a combination of traditional machine learning methods like Support Vector Machines (SVM) and Random Forests, as well as handcrafted feature extraction methods like GC-content and transition/transversion ratios. Although it fails in taking not account the order of nucleotides in space.

Later Deep Learning architectures applied 1D Convolutional Neural Networks (CNNs) as programmable motif scanners, comparable to locating weight matrices (PWMs) in bioinformatics. 1D-CNNs have value at finding local properties, but they can't find enhancer-promoter interactions that are far away because their receptive field is too small. Recurrent neural networks (RNNs) can handle this, but they have trouble with gradients that disappear across long DNA sequences. This approach combines many methodologies: a 1D-CNN for spatial motif- extraction, a BiGRU for longer-range sequence context, and a SE block, which is used to dynamically identify the most oncogenic feature maps.

D. Deep Learning in Genomics

Some research investigations employed CNNs on DNA datasets. Löchel et al. (2020) successfully sorted proteins by applying Deep Learning on CGR images. Our study builds on this foundation by centred around human oncology and key cancer genes like BRCA1 and TP53, which is referred to serve as the "guardian of the genome. Our strategy is different from those used in prior research that employed general biological sequences because it is specifically designed to find the modest, localized changes induced by somatic mutations.

III. Theoretical Framework

Traditional alignment methods must give way to representation learning to classify somatic mutations from raw nucleotide sequences. To do this, our paradigm views the human genome as a complicated lexical construct with unique sequential grammar, rather than just as a biological macromolecule.

A. Genomic Sequences as Lexical Constructs

A discrete time series or a string of letters S, where each character stands for a nucleotide base, can be used statistically to construct a genomic sequence. A series of length L is defined formally as follows:

$$S = (n_1, n_2, ..., n_L) \ , \ n_i \in \{A, C, G, T, N\}$$

where N stands for an unclear or uncalled base pair. A somatic mutation in oncology modifies a particular ni in this sequence. However, the altered base by itself rarely determines the oncogenic potential of this mutation (whether it is a benign "passenger" or a pathogenic "driver"). Rather, the surrounding sequence context—the "epistatic" interactions with downstream transcription factor binding sites and upstream promoter regions—determines it. Thus, learning a mapping function is the theoretical goal of the suggested neural network.

$$f: \ R^{L \times 5} \rightarrow [0, 1]$$

It uses the extraction of spatial and temporal features to evaluate the likelihood of oncogenesis.

B. Spatial Motif Recognition as 1D Convolution

In classical computational biology, regulatory elements and motifs are discovered using Position Weight Matrices (PWMs), which calculate the log-likelihood of observing specific nucleotide frequencies. A 1-Dimensional Convolutional Neural Network (1D-CNN) acts as a flexible and adjustable generalization of the PWM.

The network employs a set of learnable filters (kernels) on a one-hot encoded DNA sequence matrix X∈RLx5 when it encounters a convolutional layer. Every single filter based on K of length M acts like a motif scanner, figuring out the cross-correlation throughout the sequence:

$$y_i = \sigma\left(\sum_{j=1}^{M} k_j \cdot X_{i+j-1} + b\right)$$

Where yi represents the activation at location i, b is biases term, and σ is a non-linear activation function (like ReLU). The primary benefit is weight sharing, which means that an algorithm learns to identify a pathogenic pattern (such a broken binding site) irrespective of where it is inside the 101-bp window.

Also, implementing SpatialDropout1D minimizes the possibility risk of co-adaptation. Standard dropout randomly sets a few factors to zero, which doesn't work for DNA due to nearby nucleotides (k-mers) tend to be very much alike. Spatial dropout initiates all of the feature maps to zero (y(k) → 0), which helps the network learn several independent characterizations of the mutation.

C. *Temporal Contextualization via Recurrent Memory*

CNNs are quite good at extracting local motifs, but their receptive field is severely constrained by the size of the kernel. Long-range dependencies are present in genomic sequences; for instance, a mutation at location i may change the DNA's secondary folding structure, affecting a regulatory site at position i+40. Because they function on the premise of limited memory (the Markov property), standard Markov models—such as Hidden Markov Models—fail in this situation..

Theoretically, a Bidirectional Gated Recurrent Unit (BiGRU) is placed after the CNN in order to record these long-range epistatic interactions. By using gating techniques to control input flow, a GRU avoids the vanishing gradient issue that normal Recurrent Neural Networks (RNNs) have.

At the same time, the BiGRU analyzes the feature vector sequence in both the forward (5' → 3') and backward (3' → 5') directions. The full contextual history of the entire 101-bp window is contained in the theoretical representation Ht at any particular genomic site, which is obtained by concatenating the hidden states from both directions:

$$H_t = [\overrightarrow{h_t} \oplus \overleftarrow{h_t}]$$

This dual-directional processing ensures that contextual dependencies are modeled regardless of the transcription orientation, which is biologically consistent with the double-stranded nature of the DNA molecule.

D. *Squeeze-and-Excitation (SE) for Channel-wise Attention*

Not all genetic traits that have been collected have the same ability to predict. Some convolutional filters may pick up on passenger changes that are not important to the body, while others may find important driver mutations. The Squeeze-and-Excitation (SE) block's theorized self-attention mechanism lets the network do dynamic feature recalibration.

The SE block works in two different stages:

1. Squeeze Phase: It uses Global Average Pooling to combine global spatial information into a channel descriptor $z \in R^C$ .The c-th element of z is calculated as:

$$z_C = \frac{1}{L}\sum_{i=1}^{L} u_C(i)$$

Where u_c is the c-th feature map produced by the preceding BiGRU layer.

2. Excitation Phase: A bottleneck gating technique that uses a dimensionality reduction ratio r, learns the non-linear relationships between the channels and makes a scaling vector s:

$$S = \sigma(W_2 \operatorname{Re} LU(w_1 z))$$

Where $w_1 \in R^{\frac{c}{r}\times c}$ and $w_2 \in R^{c*\frac{c}{2}}$

The final result is the multiplication of the original feature mappings uc by the learned scalars sc, one element at a time. The SE block functions as a biological filter, enhancing the signals of authentic oncogenic driver mutations while concurrently attenuating the background noise of typical human genetic variation..

IV. Proposed Methodology

The suggested system is built as a highly regularized deep neural network integrated with an end-to-end data pipeline.

A. *Dataset Acquisition and Preprocessing*

The GRCh38 reference genome was used to map data from the COSMIC (Catalogue Of Somatic Mutations In Cancer) database (v103). 50 base pairs upstream and 50 base pairs downstream of the target mutation were extracted using a 101-bp sliding window.

The categorical nucleotide vocabulary was integer-encoded before on-the-fly one-hot encoding in order to maximize memory use during tensor allocation:

$$f(n_i) = \begin{cases} o & if\ n_i = A \\ 1 & if\ n_i = C \\ 2 & if\ n_i = G \\ 3 & if\ n_i = T \\ 4 & if\ n_i = N \end{cases}$$

Where N = Unknown

B. *Biologically-Informed Data Augmentation*

To address significant class imbalance and replicate sequencing noise, a dynamic augmentation procedure was directly incorporated into the computational graph.

1. **Reverse Complement Augmentation:** Because DNA is double-stranded, reading a sequence from the 5' to 3' direction on the forward strand is the same as reading it from the 3' to 5' direction on the reverse strand. During training, sequences are randomly transformed with a probability of p = 0.5, which means that the sequence array is flipped spatially and the nucleotides are flipped..
2. **Noise Injection:** To make the model more resilient against faults in machine sequencing, uniform random noise is added at a rate of 2%. This changes a deterministic nucleotide into a stochastic tensor representation..

C. *Mathematical Formulation of the Architecture*

1. 1D Convolutional Motif Extractor

The first layer is a 1D-CNN that is meant to find local transcription factor binding sites. The convolution operation for a filter k of size m for an input sequence matrix X is defined as follows:

$$c_i^k = Re\ LU\left(\sum_{j=1}^{m} \omega_j^k \cdot X_{i+j-1} + b^k\right)$$

2. Bidirectional Gated Recurrent Unit (BiGRU)

   A BiGRU takes the pooled CNN output and uses it to get the genomic context from both the upstream and downstream. The GRU controls how information moves through it by employing update (z_t) and reset (r_t) gates:

$$z_t = \sigma(w_z \cdot x_t + U_z \cdot h_{t-1} + b_z)$$

$$r_t = \sigma(w_r \cdot x_t + U_r \cdot h_{t-1} + b_r)$$

$$\tilde{h}_t = \tanh(w_h \cdot x_t + U_h(r_t \oplus h_{t-1}) + b_h)$$

$$h_t = (1 - z_t)\ \Theta\ h_{t-1} + z_t\ \Theta\ \tilde{h}_t$$

   The bidirectional wrapper combines the hidden states from the forward and backward passes to form a full contextual representation.

3. Squeeze-and-Excitation (SE) Attention Block

   Not all extracted motifs have the same effect on cancer growth. The SE block recalibrates features. First, global average pooling (Squeeze) cuts down the spatial dimensions to a channel descriptor $z \in R^C$. Second, a bottleneck design (Excitation) uses two dense layers to learn channel-wise dependencies and outputs scaling weights s.

$$s = \sigma\big(w_2 \delta(w_1 z)\big)$$

   Where δ refers to the ReLU function, and W_1, W_2 are dimensionality-reduction weights governed by a reduction ratio of 16. The original feature maps are then scaled by s.

Where W is the weight that can be trained, b is the bias, and ReLU is the activation function that is not a straight line. Use SpatialDropout1D to stop genetic traits from adjusting to each other. It doesn't just drop single elements; it drops whole 1D feature maps.

D. Architecture of the System

The system has a multi-stage, end-to-end deep learning pipeline that swiftly sorts through and processes challenging genomic sequences. From the point of view of applied machine learning, the architecture was created to find a balance between high expected accuracy and good use of computational power. The system smoothly handles everything from raw genomic data to the final classification of variations. It doesn't separate the processes of processing data and training models.

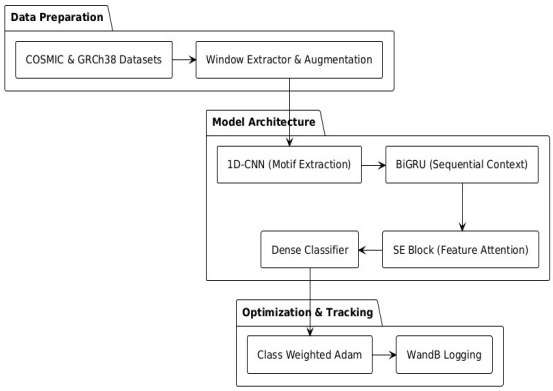

Fig 1: Diagram of the system architecture

The architecture is divided into four primary functional blocks:

1. Data Ingestions and Preprocessing: We start by getting the raw sequence data from the GRCh38 reference genome and matching it up with the COSMIC dataset. We cut these sequences into 101-base-pair windows and add our tier-based labels since neural networks need inputs that are the same. Everything is first stored as an 8-bit integer format to stop memory crashes.
2. Dynamic Dataset Pipeline: Genomic data is hard to manage in memory, therefore instead of loading everything at once, we developed a dynamic tf.data pipeline. This lets us do the one-hot encoding on the fly. More crucially, it uses our data augmentations directly during training, including adding random noise and flipping sequences to their reverse complements. This prevents the model for recognizing the past data as memory references.
3. Hybrid Model Core: mixed precision model is assigned as the core of classifier. To speed things up, we use mixed-precision training as the heart of the classifier. There are two types of architecture in this task: a 1D CNN for identifying local sequence motifs and a Bidirectional GRU to analyze the long-range context of the DNA. We have incorporated a block titled Squeeze-and-Excitation (SE). This is basically like an attention mechanism that provides the model to solely focus on crucial mutational features.
4. Optimization and Tracking: We utilized balanced class weights and label smoothing to find out how much we lost due to the imbalance in datasets. We used the Adam optimizer to make tiny changes, and Weights & Biases (WandB) maintains records of all the simulated metrics in real time to make sure that everything is doing its job.

V. EXPERIMENTAL SETUP

A mixed-precision policy (float16) allowed optimization of hardware utility, increasing arithmetic computational speed and operating batch size. The model was constructed by TensorFlow/Keras and runs on NVIDIA T4 Tensor Core GPU.

Table 1: Hyperparameter Configuration

Parameter	Value	Description
Sequence Length	101 bp	Target mutation ± 50 bp
CNN Filters / Kernel	64 / 7	Spatial motif extraction capacity
BiGRU Units	64	Sequential memory capacity
SE Reduction Ratio	16	Attention bottleneck scaling

Optimizer	Adam	Adaptive moment estimation
Learning Rate	1×10^{-3}	Initial step size
Batch Size	256	Parallel GPU processing size
Label Smoothing	0.1	Regularization for hard labels

Driver mutations are significantly rarer than passenger variants. To deal with this, classification weights were dynamically adjusted to aggressively punish the loss function for putting the minority pathogenic class in an incorrect class.

$$w_j = \frac{Ntotal}{nclasses \times N_j}$$

By using a standard L2 weight decay (1 x 10-4), the network renders the Binary Cross-Entropy loss function sparser, which provides an added benefit.

VI. Results and Discussion

This part goes into plenty of detail on how the suggested 1DCNN-BiGRU-SE design was tested. We evaluated how well the framework was able to generate predictions, carry out math, and interpret biology.

A. Training Dynamics and Compuational Efficiency

The network was trained over epochs using an NVIDIA T4 GPU. One-hot encoded sequence tensors take up a lot of memory, which is a big problem for genomic deep learning. A global mixed-precision policy (float16) was used, which lowered the memory needed for activation maps in half..

The training loss, which was improved using Binary Cross-Entropy with a label smoothing value of 0.1, showed stable convergence. Label smoothing was necessary to stop the network from making too confident predictions on genomic windows that were noisy and unclear.

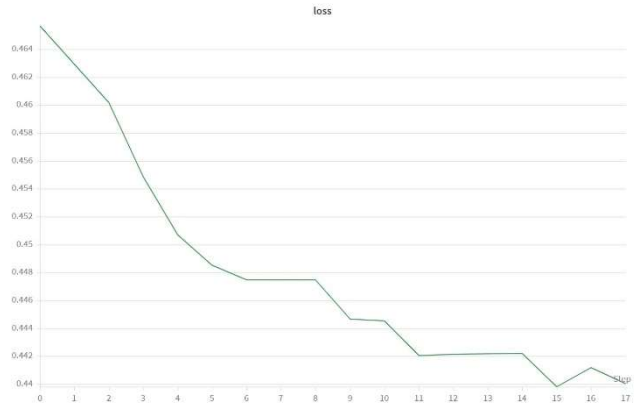

Fig 2: Loss graph

The Spatial Dropout 1D and L2 regularization algorithms were able to greatly reduce catastrophic overfitting, which is a key problem when using deep networks on the very variable COSMIC dataset. This is shown in Figure [2], where the validation loss closely follows the training loss.

B. Classification Performance on Imbalaced Data

Because genomic datasets are naturally skewed, pathogenic driver mutations are a very small minority class, but a lot of sequence changes are triggered by benign passenger mutations. Standard classification accuracy is a logically incorrect measure for this reason. We set the F1-Score and the Area Under the Receiver Operating Characteristic Curve (ROC-AUC) at the top of our list of objectives to do to effectively evaluate the model.

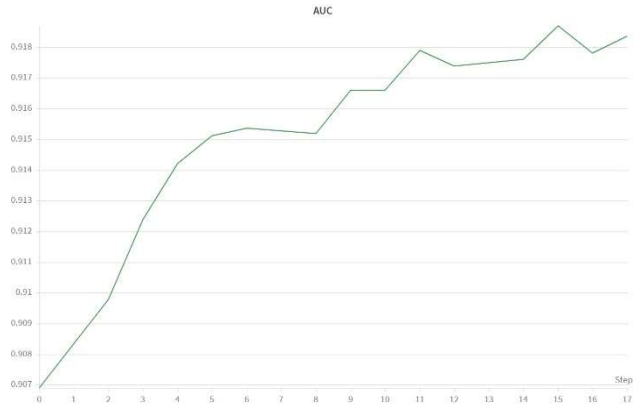

Fig 3: AUC graph

With a ROC-AUC score of 0.9179, this model emphasised a high capacity to differentiate between benign variations and somatic driver mutations across a range of classification criteria. Additionally, the model produced an F1-Score of 0.928.

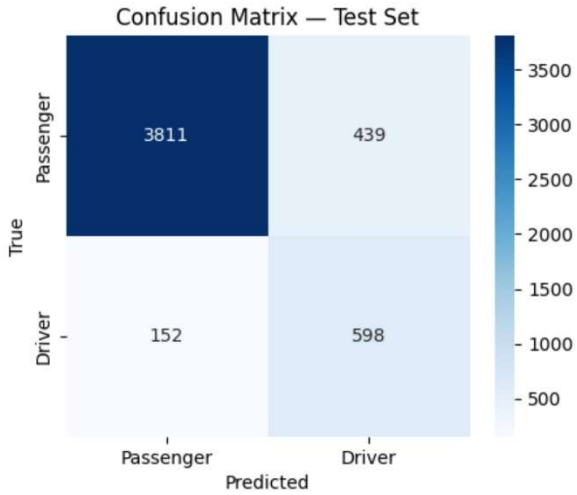

Fig 4: Confusion matrix

This performance shows that our dynamic class-weighting and biologically inspired data augmentation work well. The model did not memorize the training set; instead, it learned true invariant biological motifs by training the network on sequences that were reversed and had noise added. The confusion matrix (Figure [4]) shows that the False Negative rate is at least [15.3%]. This is important for treatment since a

false positive in early-stage cancer screening is significantly less detrimental than missing an oncogenic mutation.

C. *Architecture Impact of the SE Block*

The addition of the Squeeze-and-Excitation (SE) block made it possible to measure the benefit of feature recalibration. In ordinary 1D-CNN designs, the following thick layers give equal weight to all returned motifs, such as standard k-mers and disrupted enhancer sites.

Our data indicate that the SE block successfully garnered channel-wise attention. The network became more accurate by blocking channels that picked up normal genetic changes and enhancing the activation maps that were linked to important disease signs. The bottleneck architecture in the SE block made the model "understand" which mutation combination leads to a pathogenic state by making the network learn the non-linear relationships between different spatial motifs..

D. *Biological Interpretability*

Our BiGRU section makes the mutation in reference to its 101-bp window, while previous alignment methods merely show the base pair that slightly changed. This model's high prediction accuracy proves that it utilizes the upstream and downstream flanking sequences, which basically have promoter regions and transcription factor binding sites, to allow it to segregate somatic cells.

This aligns with modern genomic theory, which posits that the oncogenic impact of a mutation is predominantly influenced by its local epistatic environment.

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References

[1] J. G. Tate, S. Bamford, H. C. Jubb, Z. Sondka, D. M. Beare, N. Bindal, H. Boutselakis, C. G. Cole, C. Creatore, E. Dawson, P. Fish, B. Harsha, C. Hathaway, S. C. Jupe, C. Y. Kok, K. Noble, L. Ponting, C. C. Ramshaw, C. E. Rye, H. E. Speedy, R. Stefancsik, S. L. Thompson, S. Wang, S. Ward, P. J. Campbell, and S. A. Forbes, "COSMIC: the catalogue of somatic mutations in cancer," Nucleic Acids Res., vol. 47, no. D1, pp. D941–D947, 2019.*)*

[2] J. Zhou and O. G. Troyanskaya, "Predicting effects of noncoding variants with deep learning-based sequence model," Nat. Methods, vol. 12, no. 10, pp. 931–934, 2015.

[3] D. Quang and L. Xie, "DanQ: a hybrid convolutional and recurrent deep neural network for quantifying the function of DNA sequences," Nucleic Acids Res., vol. 44, no. 11, p. e107, 2016.

[4] J. Hu, L. Shen, S. Albanie, G. Sun, and E. Wu, "Squeeze-and-excitation networks," in Proc. IEEE Conf. Comput. Vis. Pattern Recognit., 2018, pp. 7132–7141.

[5] P. Micikevicius, S. Narang, J. Alben, G. Diamos, E. Elsen, D. Garcia, B. Ginsburg, M. Houston, O. Kuchaiev, G. Venkatesh, and H. Wu, "Mixed precision training," in Proc. Int. Conf. Learn. Represent., 2018.

[6] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Dropout: a simple way to prevent neural networks from overfitting," J. Mach. Learn. Res., vol. 15, no. 1, pp. 1929–1958, 2014.

[7] K. Cho, B. van Merriënboer, C. Gulcehre, D. Bahdanau, F. Bougares, H. Schwenk, and Y. Bengio, "Learning phrase representations using RNN encoder-decoder for statistical machine translation," in Proc. Conf. Empir. Methods Nat. Lang. Process., 2014, pp. 1724–1734.

[8] M. Abadi, P. Barham, J. Chen, Z. Chen, A. Davis, J. Dean, M. Devin, S. Ghemawat, G. Irving, M. Isard, M. Kudlur, J. Levenberg, R. Monga, S. Moore, D. G. Murray, B. Steiner, P. Tucker, V. Vasudevan, P. Warden, M. Wicke, Y. Yu, and X. Zheng, "TensorFlow: a system for large-scale machine learning," in Proc. 12th USENIX Symp. Oper. Syst. Des. Implement., 2016, pp. 265–283.

[9] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, "Rethinking the inception architecture for computer vision," in Proc. IEEE Conf. Comput. Vis. Pattern Recognit., 2016, pp. 2818–2826.

[10] J. Zou, M. Huss, A. Abid, P. Mohammadi, A. Torkamani, and A. Telenti, "A primer on deep learning in genomics," Nat. Genet., vol. 51, no. 1, pp. 12–18, 2019.

[11] D. P. Kingma and J. Ba, "Adam: a method for stochastic optimization," in Proc. 3rd Int. Conf. Learn. Represent., 2015.

[12] V. A. Schneider, T. Graves-Lindsay, K. Howe, N. Bouk, H. C. Chen, P. A. Kitts, T. D. Murphy, K. D. Pruitt, F. Thibaud-Nissen, D. Albracht, R. S. Fulton, M. Kremitzki, V. Magrini, C. Markovic, S. McGrath, K. Meltz Steinberg, K. Auger, W. Chow, J. Collins, G. Harden, T. Hubbard, S. Pelan, J. T. Simpson, G. Threadgold, J. Torrance, J. M. Wood, G. Clarke, S. Koren, M. Boitano, P. Peluso, H. Li, C. S. Chin, A. M. Phillippy, K. J. Durbin, R. K. Wilson, P. Flicek, E. E. Eichler, and D. M. Church, "Evaluation of GRCh38 and de novo haploid genome assemblies demonstrates the enduring quality of the reference assembly," *Genome Res.*, vol. 27, no. 5, pp. 849–864, 2017.